

APV

Funding & Issue Cost

Sources of Fund	Amount	Issue Cost	Amount of issue cost	Gross Amount
Internal Fund	6,000,000	0		6,000,000
Rights Issue	10,000,000	4%	416,666.67	10,416,667
Long term Loan	5,000,000	1%	50,505.05	5,050,505
Subsidiesd Loan	4,000,000	0		4,000,000
Total Fund	25,000,000			25,467,172

Financing Side Effects : PV of Tax Relief on Interest

	Amount	Rate	Interest	Tax Relief @ 30%	DF @ Rf of 5.5%	PV of tax releif on interest
Loan	5,050,505	8%	404,040	121,212.12	7.538	913,697
Subsidized Loan	4,000,000	6%	240,000	72,000.00	7.538	542,736

Financing Side Effects : PV of After-tax subsidy benefit

	Amount	Interest Saved	Interest saved amount	Tax shield lost @30%	After-Tax Benefit	DF @ Rf of 5.5%	PV of After Tax Subsidy benefit
Subsidized Loan	4,000,000	2%	80,000	24,000	56,000	7.538	422,128

Calculating Market Value and YTM of Bonds

Finding YTM - based on the IRR of the Cash Flows

Year	Particulars	Cash Flows
0	Market Value	-109.92
1	Interest (1-t)	5.63
2	Interest (1-t)	5.63
3	Interest (1-t)	5.63
4	Interest (1-t) + Redemption	113.63
Yield to Maturity / Gross Redemption		4.71%

Yield = Copoun Rate / MKT Value of Bond

Bond	Gov Annual Yield Curve	Credit Spread	Co. Annual Yield Curve
G	4%	0.56%	4.56%
Th	4.30%	0.78%	5.08%
De	4.70%	1.06%	5.76%
Ro	5.20%	1.35%	6.55%

Value of new bonds based on annual yield curve

Year	CF	DF	DCF
1	7.5	0.9564	7.17
2	7.5	0.9056	6.79
3	7.5	0.8453	6.34
4	115.5	0.7759	89.61
Market Value of the Bond			109.92

Currency Risk Management

Matching / Netting Off	Ads:
Owed By (Receivables)	Reduces the number of transactions.
Owed To (Payables)	Reduces the transaction cost.
	Dis Ads:
	Some Jurisdiction does not allow netting.
	May result in loss of benefit for a particulat group unit.

Forward Exc. Contracts
Ads:
Sample and easy to understand.
Dis Ads:
Forward markets does not exist for the so-called exotic currencies.
Binding Contract - you have to excersice it even if circumstances change.
Eliminates exposure to upside as well as downside movements.

Currency Futures
It is same as Forward Exchange Contracts with the following Key Differences:
* <i>Futures are for standardized amounts</i> * <i>Futures are tradable</i> * <i>Settelment can be done before expiry date</i>

Currency Options
For Payments Options Hedge
Total Payments = Expected Payments in L.C using excersize price + Un hedged payments to be made + Premium Payable
Premium = No. of Contracts * Contract Size * Premium amount --> Ex at Spot
Un-hedged =Expected amount in F.C - (No. of contracts * Contract Size * Exercise Price) -> Ex @ Forw

Interest Rate Risk Management

Futures

Strategy -- > B: SN, BL

- Step 1: Find Number of Contracts
- (Borrowe or Deposit amount/Contract Size) * (Time period/3)
- Step 2: Find Basis Risk
- (Spot - LIBOR - Future Price)

Basis Risk : Is the difference between future and spot prices, when future contract does not have predictable relationship with the spot price of instrument.

- Step 3: Finding Unused Basis Risk
- Basis Risk * (Extra Monhts/ Total Months)
- Step 4: Future Close out price
- (Spot - LIBOR) +/- Unused Basis Risk
- Step 5: Finding Tick Size
- (Contract Size/10,000) * (3/12)
- Step 6: Finding Outcome of the hedge

Actual cost / Actual Return	
(Amount * Interest * Time)	X
(Diff in future price - lock in rate) * Spot*No.of contract * Tick Size	X
Net Cost / Net Return	X

- Step 7 : Effective Interest Rate
- (Net cost or Return / Borrow or Deposit amount) * (12/Time Period)

Interest Rate Options

Strategy : Borrow-Put Deposit-Call

- Excersize Price of option
- Future close out price (From Futures)
- Excersize, Take Decision (Yes/No)
- Gain in basis point if Yes
- Total Gain (gain in basis points * No. of Contracts * Tick Size)
- Actual cost on borrowing / Actual return on deposit (From Futures)
- Premium Payable (Premium Rate * Spot * No. of Contracts * Tick Size)
- Net Cost / Net Return
- Effective Rate (calculate same as Futures)

Interest Swaps

	Main Co.	Counter Party
Record the outflow of loan Taken	-0.04	(L+0.4%)
Cancel Floating Rate First	(L+0.4%)	L+0.4%
Cancel Fixed Rate	0.05	-0.05
Net Effect	(L-0.6%)	-0.05
Charge Bank Fee	-0.003	-0.002
Effective Rate of Swap	(L+0.9%)	-0.052

Ads:

They can be arranged at any size.
Swaps make use of the principle of comparative advantage.

DisAds:

Swaps are subject to counterparty risk, they might default on arrangement.
Main company cannot take advantage of favourable interest rate changes.
Swaps are not traded.

Forward Rate Agreements

Actual Cost/Return	X
Add/Less: Gain/ Loss due o FRA	
(Libor-Forward rate)*amount*Time	X/(X)
Net Cost or Return	X

Mergers & Acquisitions

Dividend Valuation Model

The value of a company is equal to the **PV of all future dividend discounted @ Ke**

$$market\ value = \frac{D(1 + g)}{ke - g}$$

D = Current dividend

Assumptions

Dividend will increase with consant growth rate for foreseeable future.
Required return (ke) will remain constant for the foreseeable future.

Bootstraping PAT Method

PAT of Buyer
Add: PAT of Seller
Add: Post tax Synergy benefit
Combined PAT
* PE Ratio of Buyer
Value of Combined Co.

*Note P.E Ratio = Share Price/EPS

Add together method

Pre Acquisition Value of Buyer
Add: Pre Acquisition Value of Seller
Add: P.V of Synergy benefit
Value of Combined Co.

Capitalization Method

Value = No. of shares * Current Share Price

CIV Model

Step 1: Find the proxy company ROCE.

$$\text{P.V of all intangibles} = \frac{\text{Excess Earnings (1-T)}}{\text{WACC}}$$

Total Value = Capital Employed + P.V of Intangibles

Step 2:

PBIT of Target Company

Less : Capital Employed of target * Industry ROCE

Excess earnings

Less : Tax

After-tax Excess Earnings or value surplus

Creating Synergies

1 - Revenue Synergies:

Revenue synergy exists when the acquisition of the target company will result in higher revenues for the acquiring company, higher return on equity or a longer period of growth. Arises from **(1) Increased market power** **(2) Marketing synergies** **(3) Strategic synergies**

2 - Cost Synergies:

A cost synergy will result from economies of scale when the economies of scale increases the marginal cost fall and this results in cost synergies.

3 - Financial Synergies:

A Financial Synergy results from financial factors such as Tax Savings, Diversification, Cash Slack, Debt Capacity and etc.

Calculating % gain under each consideration Method

Cash offer

Current Share price	X
Cash offer	(X)
% gain	X

Share for share exchange

Current Share price	X
Combined co, share price	(X)
% gain	X

*Note: Combined co share price can be found through valuation methods.

Bonds offer

Current Share price	X
MV of bond	(X)
% gain	X

*Note: If MV of bond is not given it is found by PV of outflows (Interest+Redemption) discounted at Kd

The Greeks in BSOP:

1 - Gamma : Measures change in option value due to change in current share price.

2- Rega : Measures change in option value due to change in Volatility.

3- Theta : Measures change in option value due to change in time.

4- Rho : Measures change in option value due to Risk Free Rate.

The European Union

The EU is one of several international economic associations.

A **free trade area** exist when there is no restriction on the movement of goods and services between countries.

A **common market** encompasses the idea of a cusotms union but has a number of additional features.

The Single **European Currency**, The **Euro**.

The Customs Union

It establishes a free trade area between member states, The EU promptes free trade among member states, while acting as a **protectionist bloc** against the rest of the world.

The Single European Market

The elimination of the trade barriers directly benefit multinational companies to engage in business across the EU.

Elimination of trade restrictions

Physical barriers - customs inspection have been removed for most of goods and services.

Technical Standards - have been harmonised.

Providing **Financial Services** across the EU.

Free movement of **Capital** across the EU.

Professional Qualifications awarded in one member state is recognized in the others.

North American Free Trade Agreement (NAFTA)

It consist of 3 countries, Canada, US and Mexico.

It has formed on 1994, covering population of 360 million.

It has an economic output of \$6,000 Billions annually.

It is the second largest free trade area after EEA (European Economic Area).

World Trade Organization (WTO)

It has over 100 members, Including entire EU.

It Aims to:

Reduce the existing barriers to free trade.

Eliminate discrimination in international trade such as tariffs ans subsidies.

Acts as a **forum** for assisting free trade, helping countries negotiate and **offering a disputes settlement process**.

International Monetry Fund (IMF)

The IMF is providing financial support for countries with deficit payment balance.

The IMF can provide financial support to member countries, Most IMF loans are repayable in 3 to 5 years time.

The World Bank

The World Bank lends to creditworthy governments of developing nations to finance projects and policies that will stimulate economic development and alleviate poverty.

European Central Bank

It is responsible for administering the monetary policy of the EU Euro zone member states and is thus one of the world's most powerful central banks.

Islamic Finance

Islamic finance transactions are based on the concept of sharing risk and reward between the investor and the user of funds.

Riba

Interest or Riba is Forbidden in Islamic Finance.

Islamic Finance Contracts

Mudaraba – a partnership contract

Musharaka – a form of equity where a partnership exists and profits and losses are shared

Murabaha – a form of credit sale

Ijara – a form of lease

Sukuk – similar to a bond

Mudaraba Contract

A partnership transaction in which one party Contributes Capital and the other Contributes the Skill and Experties.

Musharaka partnership contract

It consist of atleast 2 parties, and is widely used in equity financing.

a) All partners bring a share of the capital as well as expertise to the business or project.

b) Any profits or loss will be shared to partners accoring the ratios agreed in the original contract.

Murabaha Contract

It is an instalment credit sale and is mostly used for the purchase of goods for immediate delivery on deferred payment terms.

Ijara contract

An ijara transaction is the Islamic equivalent of a lease where one party (lessor) allows another party (lessee)

to use their asset against the payment of a rental fee.

Salam Contract

It is like forward contract, good is sold today with a promise of future deliver, cash is received or paid imediately and goods later, date and quality of which are set in advance.

Istisna Contract

It is used to finance construction of projects.

Example:

If a builder wants to construct a building and does not have cash.

the builder can go for islamic bank and ask for istisna

Bank will provide funds for construction and charges profit on construction or sale of flats, payments is repaid to bank in installments.

ESG Issues (Environmental, Social and Governmental) Issues

1 - Environmental Issues: Organizations must think about the environmental impact in its decisions.

Memorize memic (**Wear Cap**)

W Water saving

E Energy effeciency

A Against green house gases

R Reforrestaion -- Plant more trees

C Climate change

A Action biodiversity loss

P Pollution Migration

2 - Social Issues: Organizations must think about the social factors in its decisions.

Memorize memic (**VERSE**)

- V Valient during crises - Alert being during crises - earthquake - Flooding - Providing food and shelter.
- E Employee health.
- R Respect diversity (gender balance, ethnic balance, race)
- S Saftey of people and processes
- E Equity in pay

3 - Governance Issues: Organizations must think about the governance issues in its decisions.

Memorize memic (**TEEP-C**)

- T Transparent Management Structure
- E Employee relation
- E Executive compensation
- P Pervert Bribery
- C Cyber Security and Privacy

Notes:

Focusing on ESG may be costly in the short term but will yeild benefits in long term.

ESG issues if not addressed may effect cash flows as well as business value. It may also result in high cost of capital.

As part of ESG, Companies and Individuals are also expected to make " Responsible Investments" and move towards "Green Finance".

Responsible Investments

Invest in companies that creat a positive impact on Environment and Society.

Shareholders of companies are also expected to push directors to invest in Environmental related projects.

Green Finance (Sustainability)

Green finance refers to financial products, services and invstments that focus on supporting green (Sustainable) Activities.

Any Activity promoting, climate change mitigation, resource effeciency, clean energy, Ecological conservation etc.

E.g: Electric Cars, Solar Panels, Trees plantation, Pollution reduction.

The purpose is to ensure that economic growth is coupled with positive environmental outcomes.

4 ways in which companies can raise green finance

1. Green loans

Loans given by commercial banks to fund environment friendly activites, for e.g renewable energy installations.

2. Sustainability linked loans

Borrowes given incentives to improve sustainability performance. Loan terms are linked to sustainability targets, if the targets are met then companies get favourable interst rates. (e.g reduce wastage).

3. Green Bonds

Debt instruments issued by government or companies to raise finance for environmental focused projects, proceeds from those bonds are specifically used to finance projects with positive environmental impact.

4. Green Funds

are investment vehicles that put together capital from multiple investors to finance green projects.

Dutch Auction

Defination:

A dutch auction is the process of selling shares by companies, the objective is similar to an IPO but the method and procedure if selling shres is very different in a duthc auction compared to IPO.

Ads:

Savings in transaction cost as no underwriter required like an IPO as dutch auction usually take place without underwriter.

Price transparency as every bidder is getting at same price.

All investores benefit when company selects lowest bid, even people willing to pay higher price get shares at low price making it beneficial for them.

Dis-Ads:

Risk of share unsubcription is there, as there is no underwriter it is possible some shares may remain unsubscribed.

Risk of share price crashing after auction, since retail investores may give bids and get the shares, they might feel later we got it at high price and may liquidate it leading to crash in share price